Institute for Advanced Studies
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Sentence-level Relation Extraction using Deep Neural Networks

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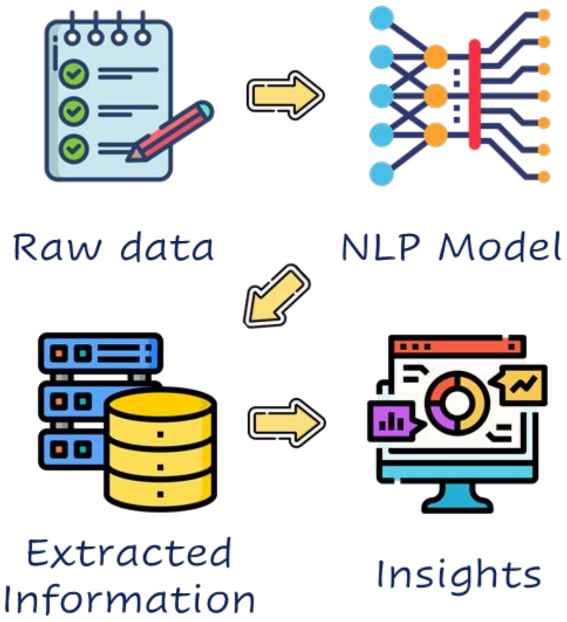
Outline

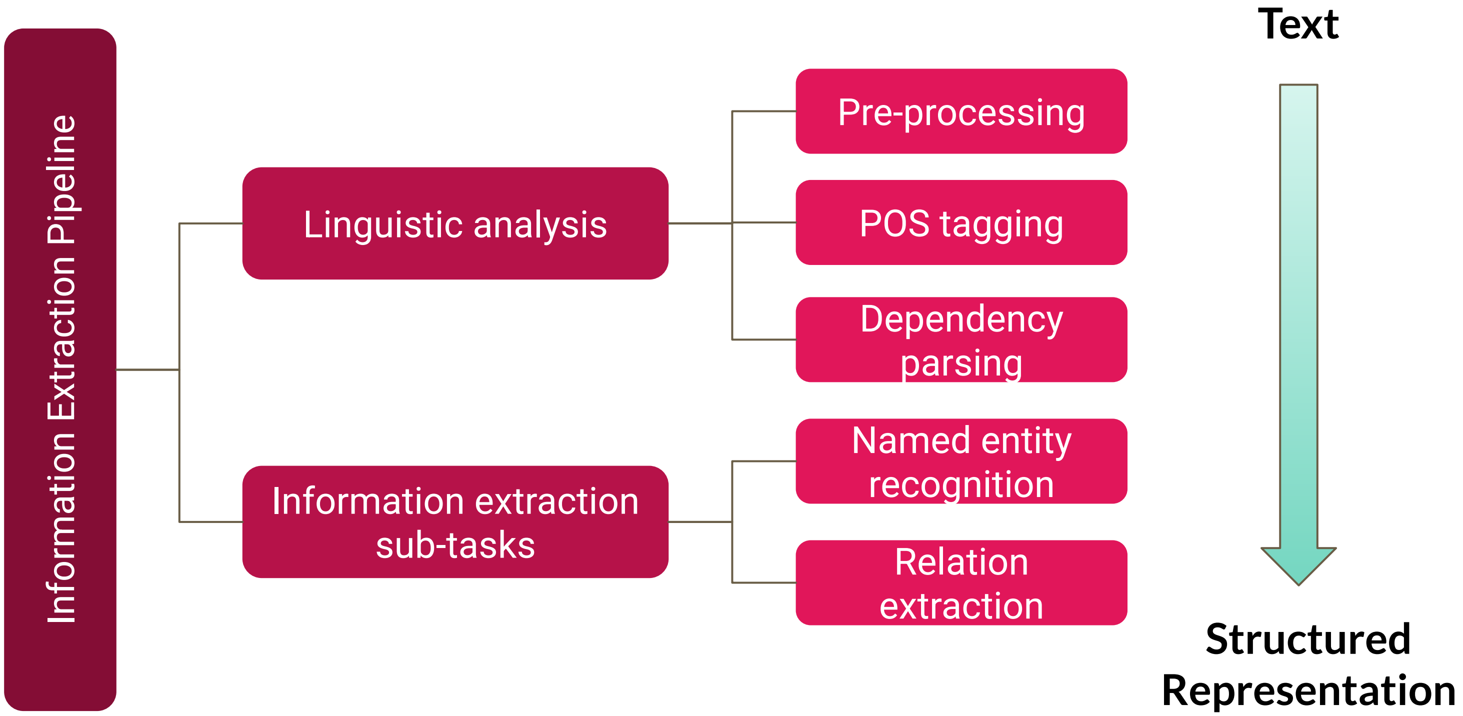
- Introduction
- Related Work
- Contribution
- Conclusion

Introduction



[https://devblogs.microsoft.com/cse/tag/named-entity-recognition/]

Information Extraction



Named Entity Recognition

Detects and assigns types or categories to entities in a text. Typically, the types are predefined and depend on the application they are used in.

Donald Trump served as the 45th president of the United States.

[person] [location]

Relation Extraction

The process of predicting and labeling semantic relationships between named entities.

President_of

Donald Trump served as the 45th president of the United States.

[person] [location]

Relation Extraction - cont.

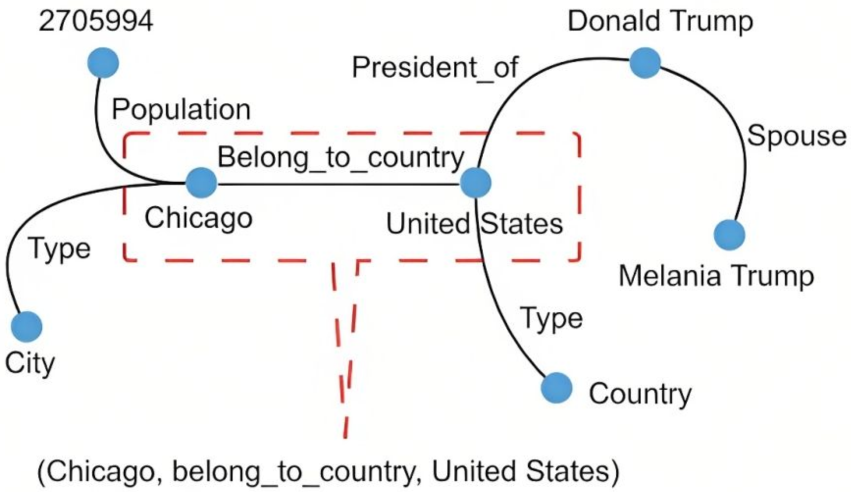
Unstructured Text Data

Chicago is located in the **United States.**

Head entity

Tail entity

Structured Representation



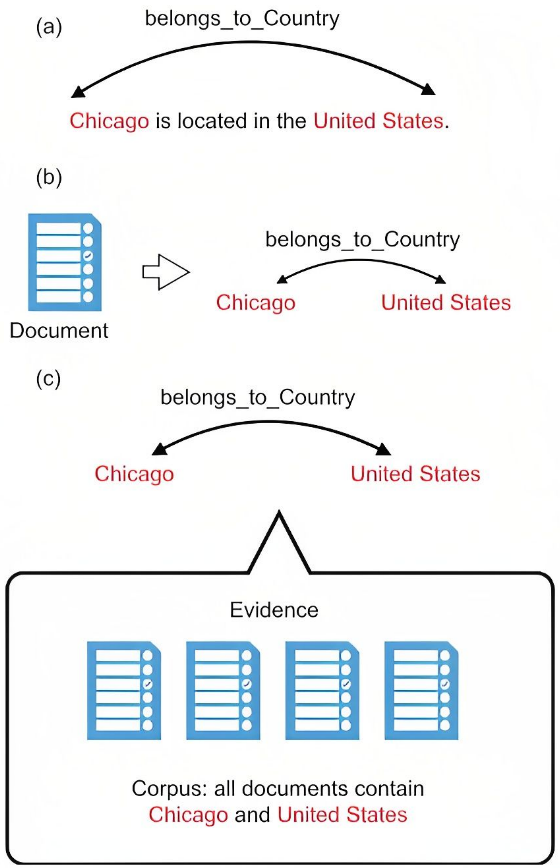
[Liu Kang - 2020]

Relation Extraction - cont.

(a) Sentence-level

(b) Document-level

(c) Corpus-level



[Liu Kang - 2020]

Relation Extraction - cont.

Supervised methods

Consider RE as a multi-class classification problem.

Relation triple = (Head entity, Relation, Tail entity)

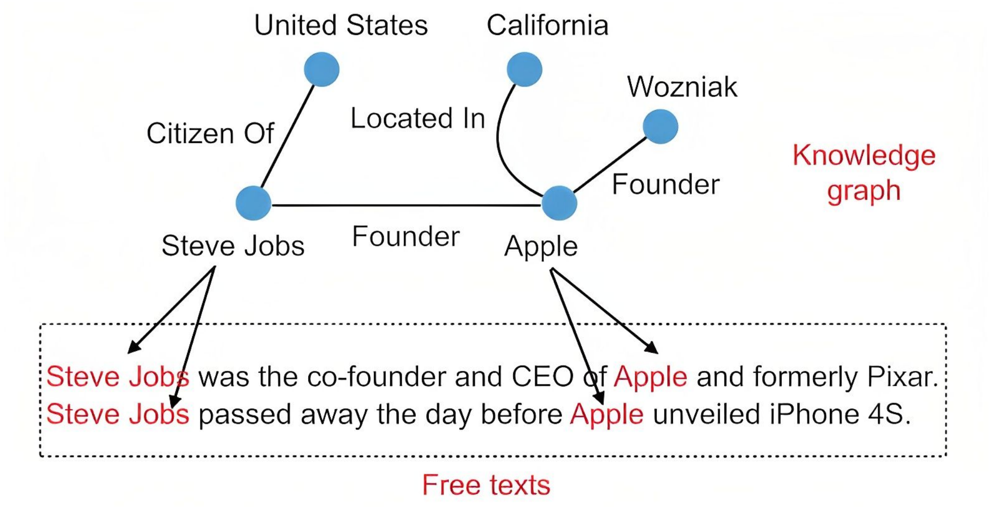
Feature vector attributes:

- Lexical features
- Dependency parse features
- Entity mention features

Relation Extraction - cont.

Distant supervision methods (weakly supervised)

Aligning relational triplets in knowledge base with free texts to automatically generate labeled training dataset.

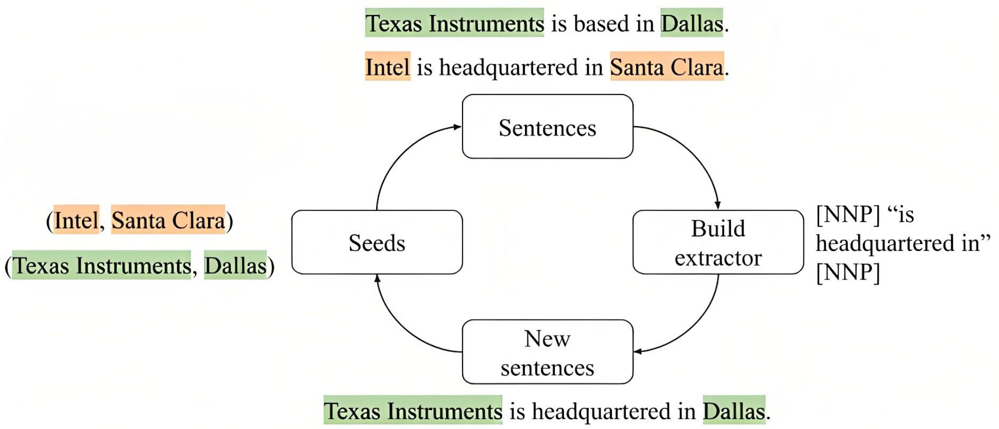


[Liu Kang - 2020]

Relation Extraction - cont.

Semi-supervised methods

Methods in this setting typically rely on **bootstrapping**, or **bootstrap learning**.



[Liu Kang - 2020]

Relation Extraction - cont.

Unsupervised

The context information of different entity pairs with the same semantic relation is relatively similar.

- Extract entity pair and context
- Cluster entity pair based on context
- Annotate semantic relation of classes

Datasets

Statistics for Relation Extraction Datasets

Dataset	Examples	Neg. examples(%)	Relations	Supervision
SemEval-2010 Task 8	10,717	17.4%	19	traditional
TACRED	106,264	79.5%	42	traditional
NYT-10	522,611	-	53	distant

Datasets - cont.

SemEval-2010 Task 8

The dataset for the SemEval-2010 Task 8 is a dataset for classification of semantic relations between pairs of nominals.

"He had chest pains and <e1>headaches</e1> from <e2>mold</e2> in the bedrooms."
Cause-Effect(e2,e1)
Comment:

Related Work

Relation Extraction using Pre-trained Language Models

- Enriching Pre-trained Language Model with Entity Information for Relation Classification **(R-BERT).** [Wu and He - 2019]
- Matching the Blanks: Distributional Similarity for Relation Learning **(MTB).** [Soares et.al - 2019]

R-BERT

Goal: Incorporate the pre-trained language model for relation classification.

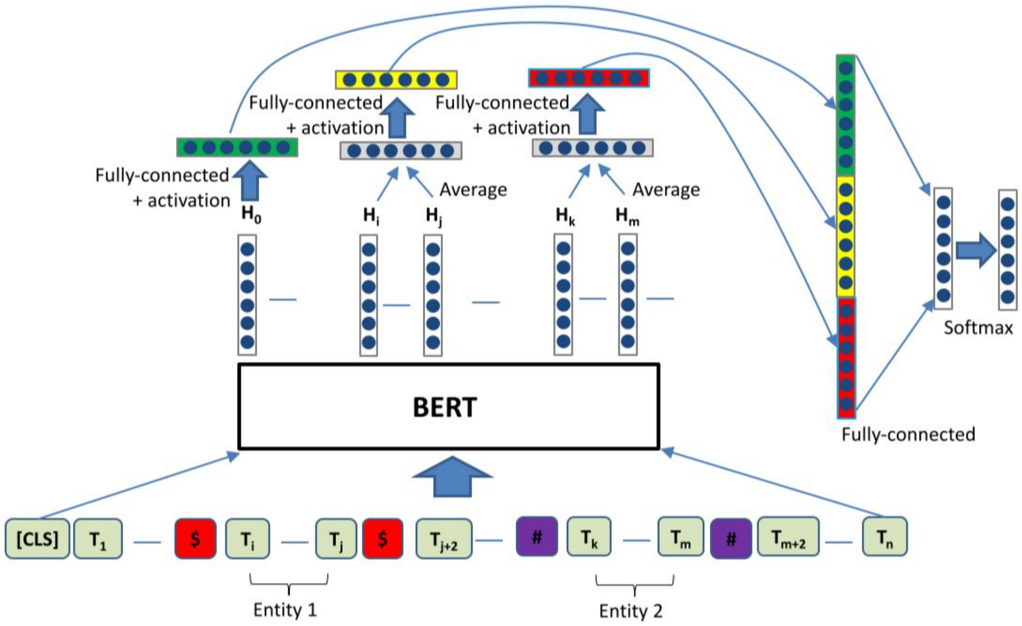
Approach: Identify the locations of the two target entities and transfer the information into the pre-trained BERT language model.

Advantages:

- increase data efficiency and thus performance in settings with limited labeled data.
- Incorporating information from the target entities to classify relation statements.

R-BERT - cont.

Model Architecture



[Wu and He - 2019]

$$H'_0 = W_0\left(tanh(H_0)\right) + b_0$$

$$H'_1 = W_1\left[tanh\left(\frac{1}{j-i+1}\sum_{t=i}^{j} H_t\right)\right] + b_1$$

$$H'_2 = W_2\left[tanh\left(\frac{1}{m-k+1}\sum_{t=k}^{m} H_t\right)\right] + b_2$$

$$h'' = W_3\left[concat\left(H'_0, H'_1, H'_2\right)\right] + b_3$$

$$p = softmax(h'')$$

MTB

Goal: Learn ontology independent relation representation.

Approach: Distant supervision through “matching the blanks” task.

Advantages:

- Use transfer learning to teach the model how to predict the relation classes.
- Group various ways of expressing the same or similar relations between entities in an embedding space.

MTB - cont.

Pre-training Data

"If two entities participate in a relation, any sentence that contain those two entities might express that relation." [Mintz et al. - 2009]

Russian forces had taken control of major cities in Ukraine's east.

Russian troops crossed Ukraine's border bring with them hundreds of tanks.

The city's British rulers and Russian refugees may be long gone but their food has remained

MTB - cont.

Pre-training Task

Key insight: Distributional similarity, Applied to relations.

$$h_r = f_\Theta(r)$$

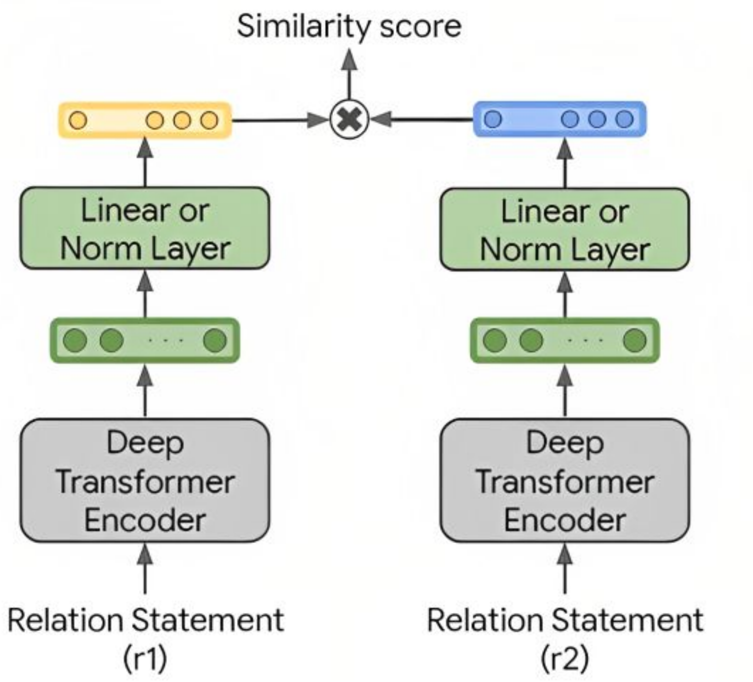
[ ] forces had taken control of major cities in [ ]'s east.

[ ] troops crossed [ ]'s border bring with them hundreds of tanks.

The city's [ ] rulers and [ ] refugees may be long gone but their food has remained

MTB - cont.

Model Architecture



$$p(l = 1|\mathbf{r}, \mathbf{r}') = \frac{1}{1 + \exp f_\theta(\mathbf{r})^\top f_\theta(\mathbf{r}')}$$

$$\mathcal{L}(\mathcal{D}) = -\frac{1}{|\mathcal{D}|^2} \sum_{(\mathbf{r},e_1,e_2)\in\mathcal{D}} \sum_{(\mathbf{r}',e_1',e_2')\in\mathcal{D}} \delta_{e_1,e_1'}\delta_{e_2,e_2'} \cdot \log p(l = 1|\mathbf{r}, \mathbf{r}') + (1 - \delta_{e_1,e_1'}\delta_{e_2,e_2'}) \cdot \log(1 - p(l = 1|\mathbf{r}, \mathbf{r}'))$$

Contribution

Contribution - 1

Transferring the pre-trained knowledge gained by completing the "matching the blanks" task to R-BERT model.

Contribution - 2

Examining how the size of a training dataset affects the performance of a relation classification task.

Contribution - 1

Goal: Transfer relation representation knowledge to the task-specific relation classification model.

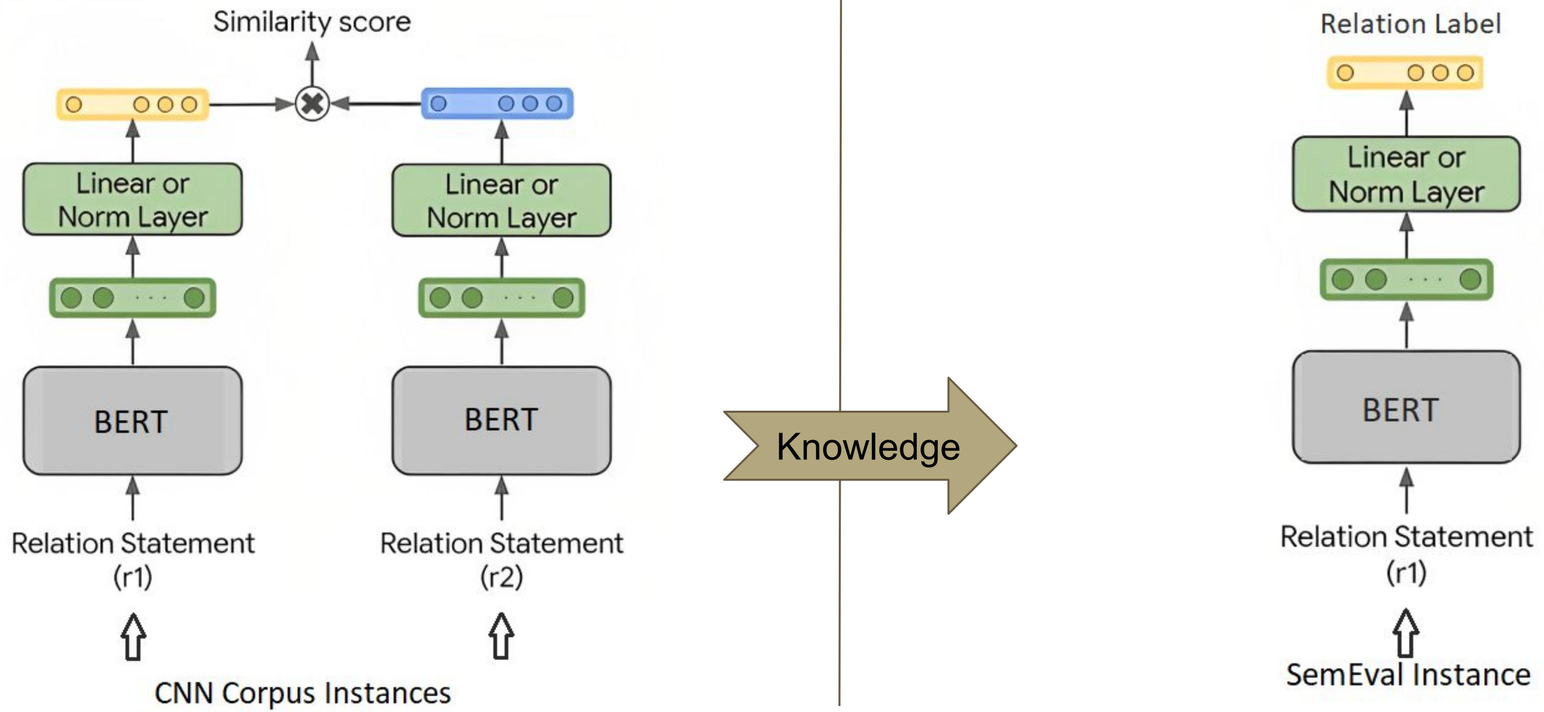
Approach: Pre-train MTB relation representation over CNN corpus, then transfer weights of pre-trained model to R-BERT model .

Results: Improvement on the results of fine-tuning the R-BERT model on the SemEval 2010 Task 8 dataset.

Contribution - 1 - cont.

Pre-training Step
(MTB model)

Fine-tuning Step
(R-BERT model)



Contribution - 1 - cont.

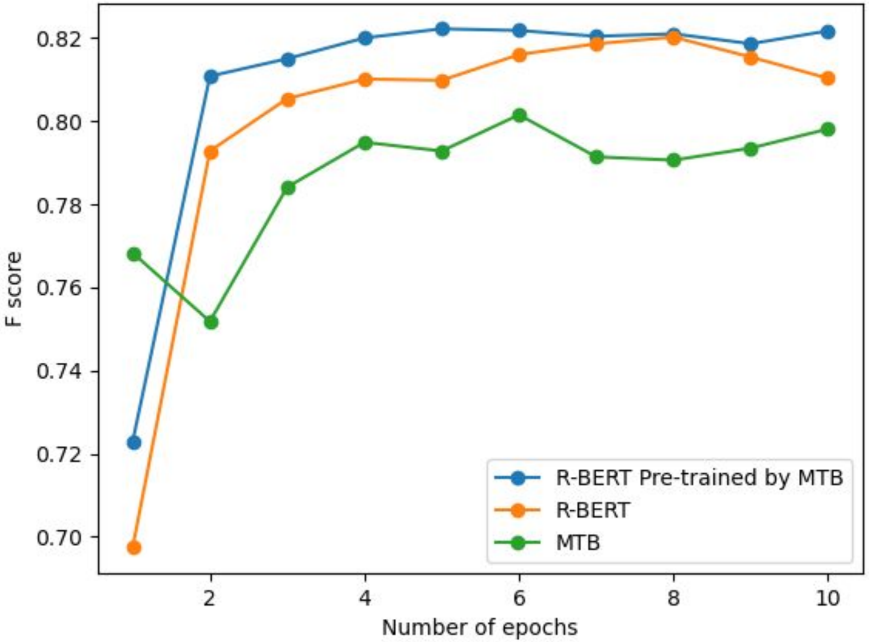
Experiments

Pre-training Model	Fine-tuning Model	Accuracy	Precision	Recall	F score
Pre-trained BERT	R-BERT	0.8381	0.7927	0.8316	0.8103
Training MTB over CNN Corpus	MTB	**0.8445**	0.7895	0.8069	0.7981
Training MTB over CNN Corpus	R-BERT	0.8425	**0.8106**	**0.8353**	**0.8217**

Model's Fine-tuning Performance after 10 epochs.

Contribution - 1 - cont.

Experiments



Contribution - 2

Goal: Impact of dataset size on relation classification models.

Approach: Perform 4 different Relation Classification models over 8 subsets of SemEval-2010 Task 8 Dataset.

Results: Impact of training dataset size on the performance of models reduce as much as the size of the training data increase regardless of what kind of classification model is used.

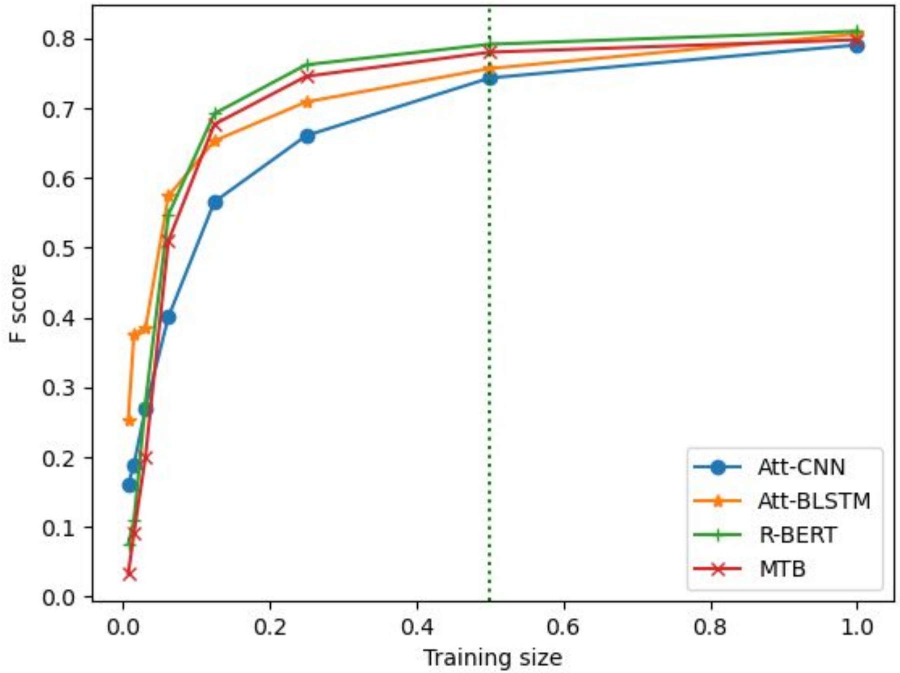
Contribution - 2 - cont.

SemEval-2010 Task 8 Subsets

Subset	Portion	Number of Samples
1	1/128	62
2	1/64	125
3	1/32	250
4	1/16	500
5	1/8	1000
6	1/4	2000
7	1/2	4000
8	1	8000

Contribution - 2 - cont.

Experiments



Conclusion

We have proposed …

- Transferring the pre-trained knowledge gained by completing the "matching the blanks" task to R-BERT model.

- Examining how the size of a training dataset affects the performance of a relation classification task.

References

[1] https://devblogs.microsoft.com/cse/tag/named-entity-recognition/

[2] Iris Hendrickx, Su Nam Kim. SemEval-2010 Task 8: Multi-Way Classification of Semantic Relations between Pairs of Nominals.

[3] Wang, Hailin, et al. "Deep neural network-based relation extraction: an overview."

[4] Alt, Christoph Benedikt. *Neural sequential transfer learning for relation extraction.*

[5] Wu, Shanchan, and Yifan He. "Enriching pre-trained language model with entity information for relation classification."

[6] Livio Baldini Soares Nicholas Arthur FitzGerald Jeffrey Ling Tom Kwiatkowski.ACL 2019

[7] Shen, Yatian, and Xuan-Jing Huang. "Attention-based convolutional neural network for semantic relation extraction."

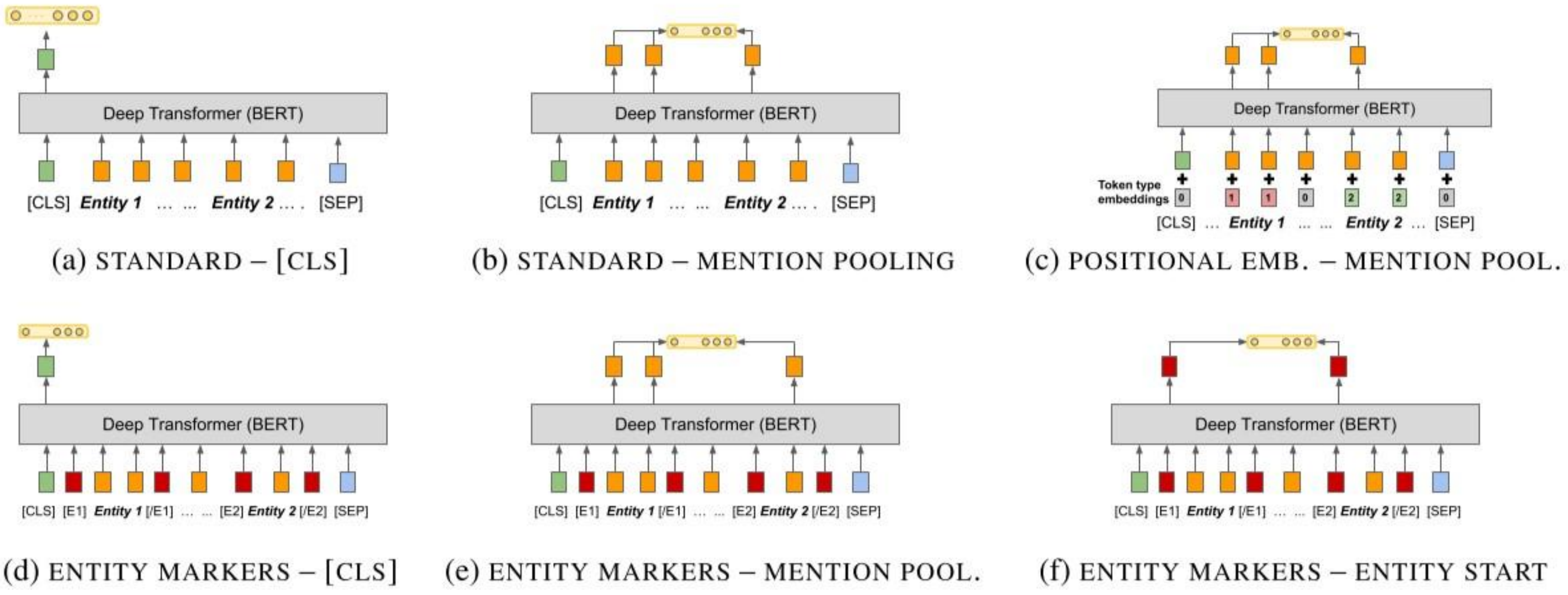
[8] Zhang, Shu, et al. "Bidirectional long short-term memory networks for relation classification." Proceedings of the 29th Pacific Asia conference on language, information and computation. 2015.

Thank you!

Any Questions ?

Appendix

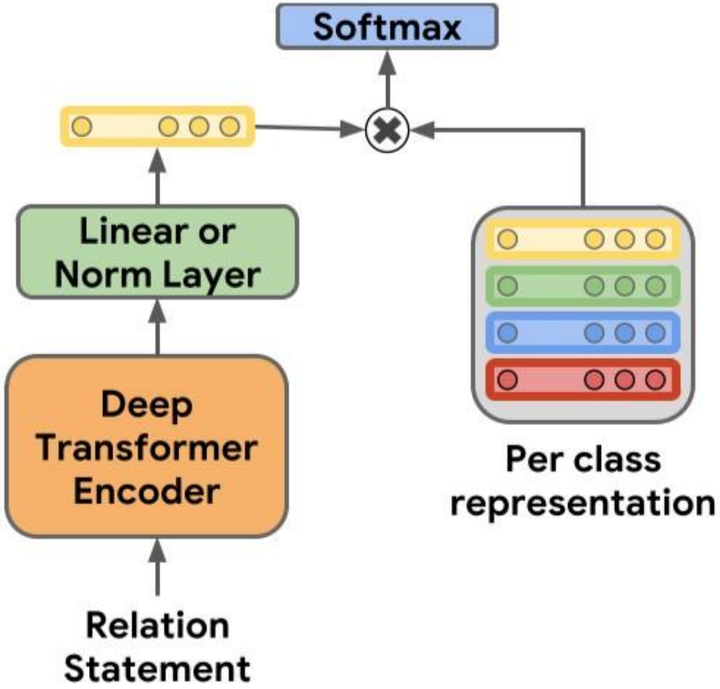
Input and output representation of MTB model



(a) STANDARD – [CLS]
(b) STANDARD – MENTION POOLING
(c) POSITIONAL EMB. – MENTION POOL.
(d) ENTITY MARKERS – [CLS]
(e) ENTITY MARKERS – MENTION POOL.
(f) ENTITY MARKERS – ENTITY START

Appendix

MTB fine-tuning



Future Work

In the pre-training step of MTB model, we are going to ...

- Apply relation learning objective over larger corpus.
    - Number of semantically similar instances for each relation will increase.

- Utilize entity type feature of pre-training data.
    - Number of false positive instances will decrease.